

Alfido Internship — Cloud Computing Domain

Task 2 Submission Document: Deploy a Virtual Machine & Install Web Server

Intern Name: J M Praveenambigai

Task Title: Deploy a Virtual Machine & Install Web Server

1. Project Goal

The objective of this task is to launch a cloud compute instance (virtual machine), configure standard network firewall routing rules for open web inspection on Port 80, install an enterprise-grade web server engine (Nginx), and overwrite the system deployment environment with a custom operational landing verification page.

2. Cloud Infrastructure & Configuration Details

- **Cloud Provider:** Amazon Web Services (AWS)
- **Core Service Used:** Amazon EC2 (Elastic Compute Cloud)
- **Operating System Distribution:** Ubuntu Server 24.04 LTS
- **Public IP Gateway Address:** <http://16.112.159.131>

3. Implementation Verification & Deliverables

Step A: Active Virtual Machine Deployment

The EC2 compute environment was successfully spun up and attached to default regional virtualization clusters.

☐	Name	Instance ID	Instance state	Instance type	Status check
☐	Praveena_task...	i-0bc823f0abfede99a	Running	t3.micro	3/3 checks passed
Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	
ap-south-2a	ec2-16-112-159-131.ap...	16.112.159.131	–	–	
Monitoring	Security group name	Key name	Launch time		
disabled	launch-wizard-1	task2-key	2026/07/11 11:00 GMT+5:30		
Launch time	Platfor...	Managed	Operator		
2026/07/11 11:00 GMT+5:30	Linux/UNIX	false	–		

Step B: Security Group Firewall Engineering (Port 80)

Inbound firewall settings were explicitly adjusted to create a routing gateway rule, letting standard HTTP requests pass over the network interface via open TCP parameters.

i-0bc823f0abfede99a (Praveena_task2_VM)

Security group rule ID	Port range	Protocol	Source
sgr-0f70c47dc0a7fc690	80	TCP	0.0.0.0/0

Step C: Secure Key-Pair Terminal Connection

Terminal authentication was executed locally from a host terminal shell utility using the asymmetric private encryption file (task2-key.pem).

```
C:\Users\prave\Downloads>ssh -i "task2-key.pem" ubuntu@16.112.159.131
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Jul 13 13:44:36 UTC 2026

System load:  0.01           Temperature:   -273.1 C
Usage of /:   39.9% of 6.61GB Processes:    122
Memory usage: 35%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.8.41

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

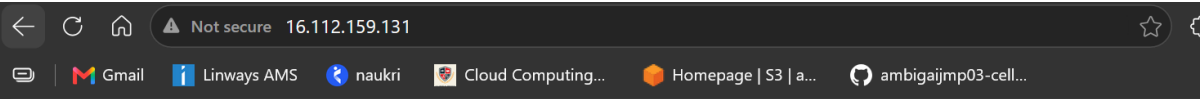
Expanded Security Maintenance for Applications is not enabled.

18 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Step D: Production Web Delivery Verification

Inputting the assigned public cloud IP mapping inside a fresh routing request returns the customized index page asset served directly by the Nginx engine.



Hello from Cloud VM